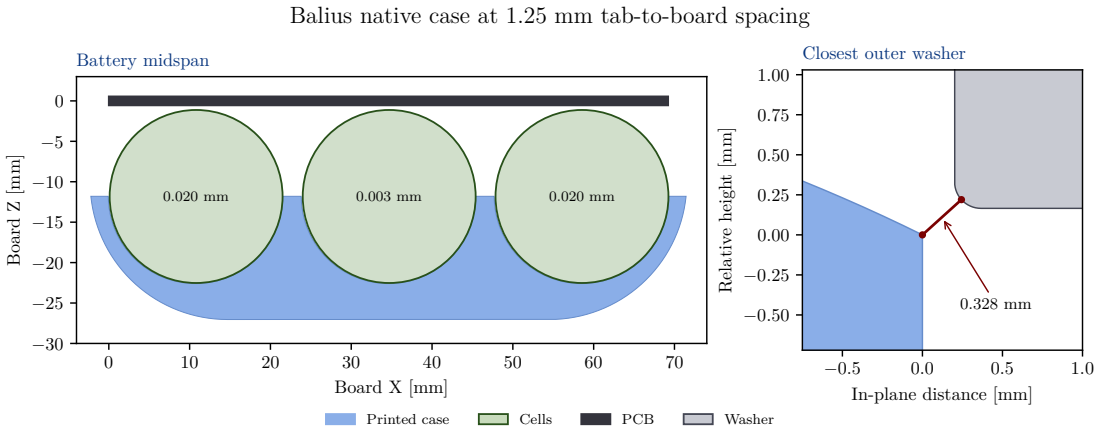
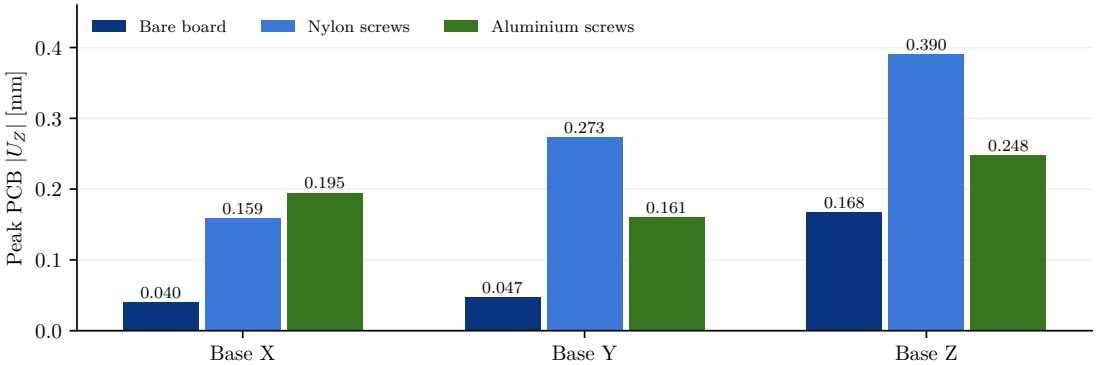
The three-pocket Balius case fits at 1.25 mm stand-off



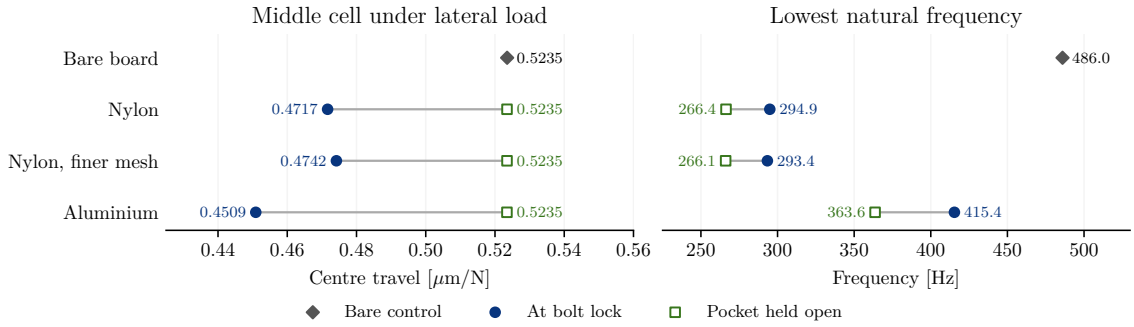
The dense PLA shell adds 73.7 g to the 251.8 g bare board. The middle-cell clearance is 0.003 mm and the closest washer clears by 0.328 mm. The rigid left cell reaches its pocket after **0.020 mm toward the middle cell** or **0.165 mm away from it**.

The snug case increases board travel in the seeded clips



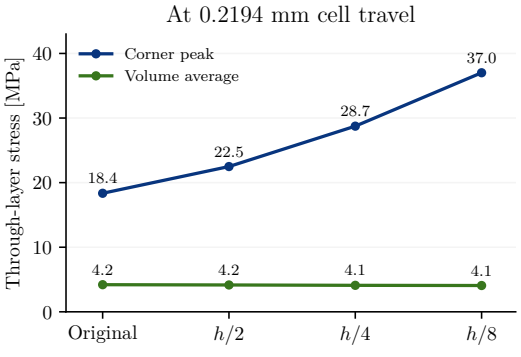
During the displayed base-Z interval, PCB travel normal to the board reached **0.168 mm bare**, **0.390 mm with nylon screws** and **0.248 mm with aluminium**. All 22 paired movie subjects are available.

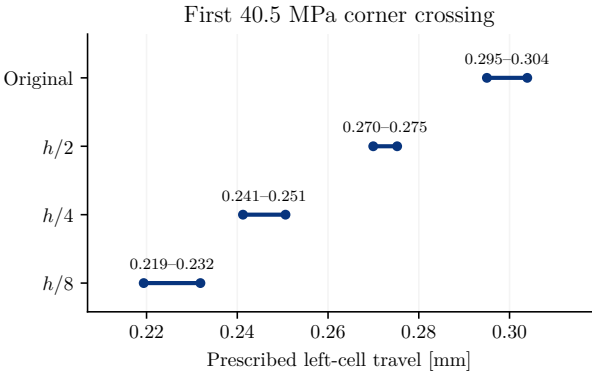
Middle-cell restraint depends on contact with the pocket



At bolt lock, the pocket took **10.4% of a small sideways load with nylon screws** and **14.5% with aluminium**. Holding the pocket open in the whole-board calculation brought cell travel within **0.011% of bare** with either mount. The lowest frequencies also fell, to **266 Hz with nylon** and **364 Hz with aluminium**. The four tabs still attached the case to the board.

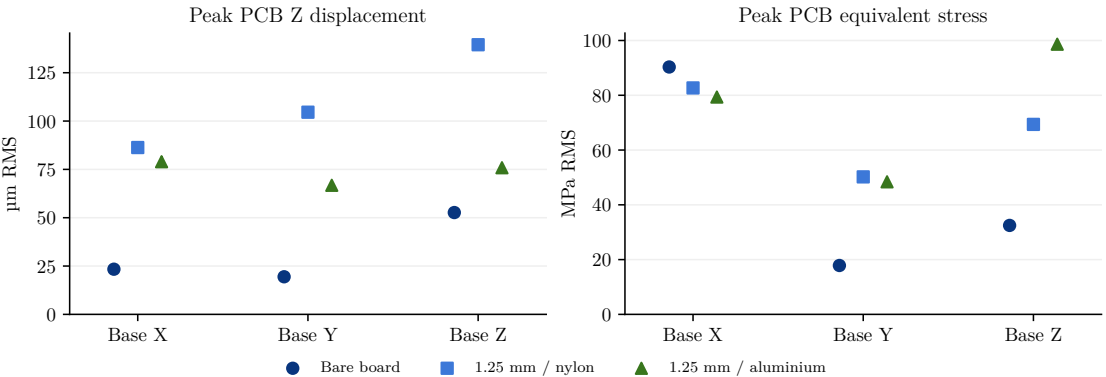
The tab-corner stress depends strongly on mesh size





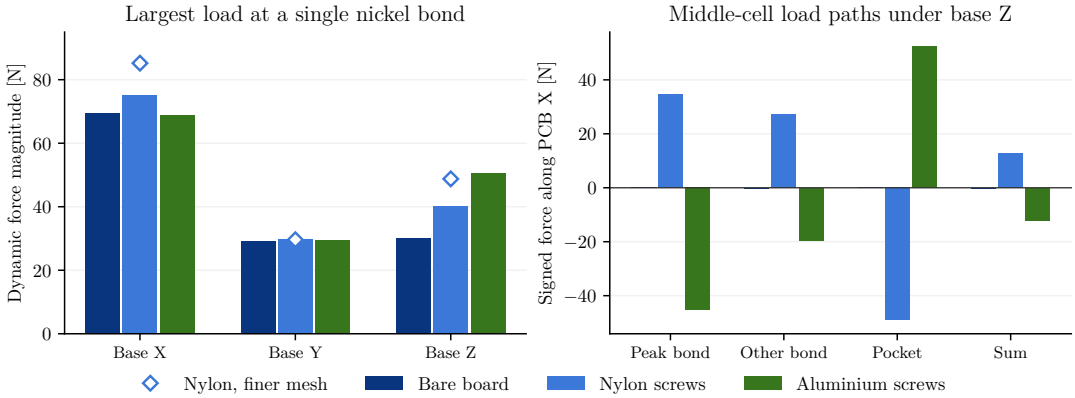
At **8.54 N**, cell travel fell from **0.500 mm bare** to **about 0.034 mm** with nylon screws. Under the same native boundary motions, local refinement moved the PLA corner crossing from about **0.30 to 0.23 mm**. The corner result has yet to converge.

Aluminium cuts board travel relative to nylon but raises base-Z stress



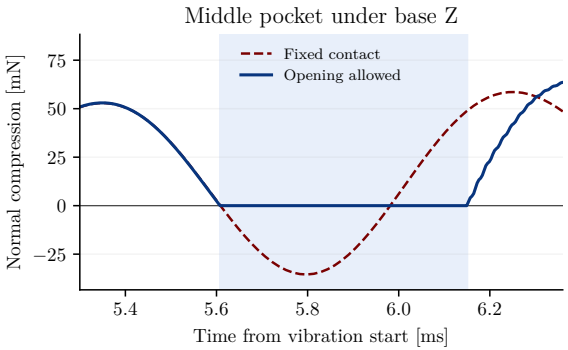
Under base Z, peak PCB stress reached **32.5 MPa RMS bare**, **69.4 with nylon** and **98.7 with aluminium**. Under base Y both mounts raised stress over nearly the entire board. These native spectra hold the installed contact state fixed.

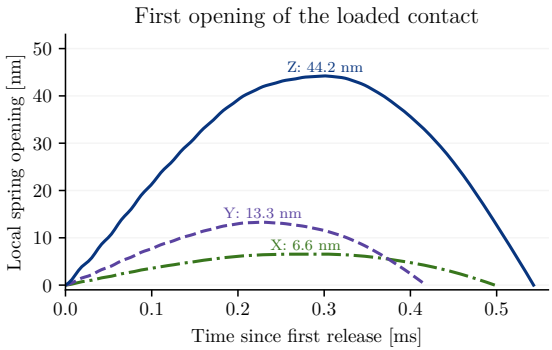
Board-normal vibration loads the middle cell's nickel bond



The largest base-Z bond load rose from **30.2 N bare** to **40.1 N with nylon screws** and **50.5 N with aluminium**. The finer nylon mesh gave **48.8 N**. All cased base-Z peaks occur at the middle cell. Bars use native histories at the installed contact state. The diamond uses independent integration on the finer native modes.

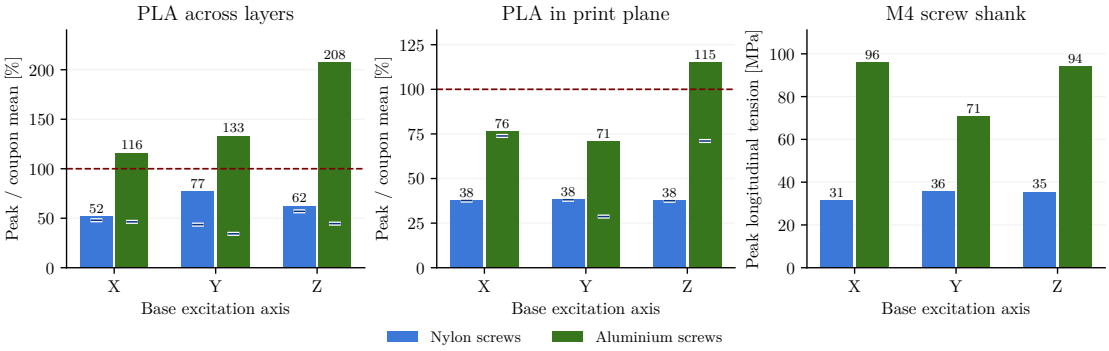
The middle cell still intersects a second PLA face at peak opening





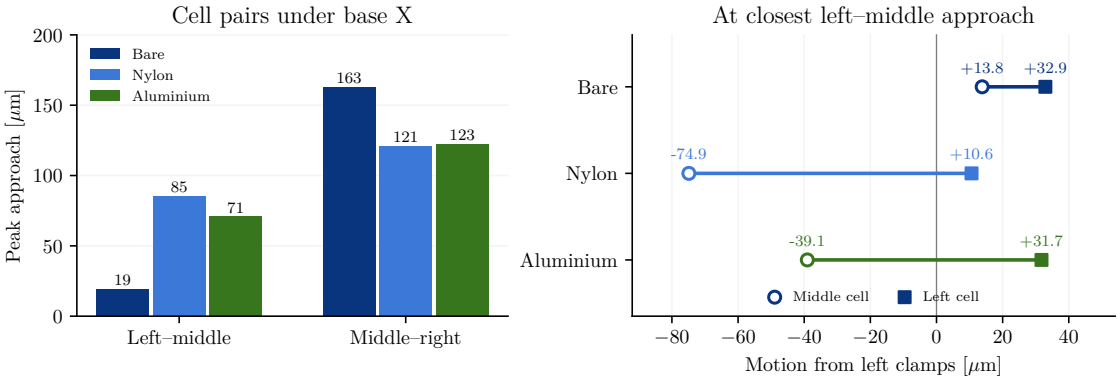
In the independent base-Z calculation, the loaded contact opened for **0.54 ms** and reached a **44 nm** gap. At peak opening, a neighbouring PLA face still intersected the middle cell. That face also intersected at native bolt lock, where ANSYS reported zero pressure on it. A matched preload will check whether surface projection transfers load there.

The printed tab exceeds its coupon reference with aluminium screws



On the nominal mesh the tab reached **77% of the through-layer coupon mean with nylon** and **208% with aluminium**. Refining the nylon case shifted its maximum to **71.5% under base X**. The aluminium mesh check is queued. Horizontal marks exclude the 2 mm bearing band.

Under base X, the left and middle cells move closer together



The left cell's peak travel relative to its clamps fell by **29% with nylon** and **12% with aluminium**. Rotation accounted for **82%** of that travel in all three assemblies. With the middle cell free to move, peak approach rose from **19 μm bare** to **85 and 71 μm**. Contact stays fixed at bolt lock in these native histories.